

The Erdős Unit Distance Problem: Incidences and Algebraic Deconstruction

Abstract

This document outlines a structural approach towards resolving the Erdős Unit Distance Problem. We formalize the axiomatic basis of point-circle incidences and propose a new decomposition lemma utilizing algebraic topology and the polynomial method.

1 Introduction and Axiomatic Definitions

The Erdős Unit Distance Problem, formulated in 1946, asks for the maximum number of unit distances that can exist among n points in the Euclidean plane. Erdős proved a lower bound of $n^{1+c/\log \log n}$ and an upper bound of $\mathcal{O}(n^{3/2})$, which was later improved by Szemerédi and Trotter, and further bounded using crossing lemma techniques.

Definition 1 (Unit Distance Graph). Let $P \subset \mathbb{R}^2$ be a finite set of points such that $|P| = n$. The unit distance graph $G = (P, E)$ is defined by setting $\{p, q\} \in E$ if and only if $\|p - q\|_2 = 1$. The total number of edges is denoted as $u(P) = |E|$.

Definition 2 (Unit Distance Function). The maximum number of unit distances for any set of n points is given by $u(n) = \max_{|P|=n} u(P)$.

Conjecture (Erdős, 1946). For any $\varepsilon > 0$, there exists a constant C_ε such that $u(n) \leq C_\varepsilon n^{1+\varepsilon}$.

2 Contextual Literature Research

The foundational result by Szemerédi and Trotter (1983) bounds the number of point-curve incidences. By treating each point as the center of a unit circle, the problem translates into bounding incidences between n points and n unit circles. Later, the application of the polynomial method by Guth and Katz (2015) to the Erdős distinct distances problem introduced new algebraic frameworks. Our approach draws an analogy with the Elekes-Sharir framework, transforming distance constraints into intersection properties of symmetric varieties in $\mathbb{R}^3$.

3 Proof Strategy and Lemmas

Our strategy involves decomposing the unit distance graph into bipartite subgraphs with controlled algebraic complexity.

Lemma 1 (Polynomial Partitioning for Circles). For any set of n points in $\mathbb{R}^2$ and an integer D , there exists a non-zero polynomial $F \in \mathbb{R}[x, y]$ of degree at most D such that $\mathbb{R}^2 \setminus Z(F)$ is partitioned into $\mathcal{O}(D^2)$ cells, each containing at most $\mathcal{O}(n/D^2)$ points.

Proof strategy: This extends the standard Guth-Katz partitioning by specifically considering the intersections with varieties representing unit circles, controlling the degrees of the intersection curves.

Lemma 2 (Incidence Bound on Algebraic Curves). The number of incidences between n points and m unit circles, under the condition that no more than C circles intersect in any common pair of points (except for the trivial maximum of 2 unit circles intersecting at 2 points), is strictly bounded by $\mathcal{O}(m^{2/3}n^{2/3} + m + n)$.

Proof strategy: We apply the incidence bounds derived from crossing numbers and Szemerédi-Trotter. Since any two distinct unit circles intersect in at most two points, the pseudo-line configuration is satisfied locally.

4 Informal Proof (Zero Ellipse)

Let us explicitly detail the application of Lemma 2. Let P be a set of n points. For each $p \in P$, define a unit circle $C_p = \{x \in \mathbb{R}^2 \mid \|x - p\|_2 = 1\}$. Let $\mathcal{C} = \{C_p \mid p \in P\}$. Thus, we have $|\mathcal{C}| = n$ unit circles. The number of unit distances $u(P)$ is exactly half the number of incidences between the points P and the circles $\mathcal{C}$. Formally, $2u(P) = I(P, \mathcal{C}) = |\{(p, C_q) \in P \times \mathcal{C} \mid p \in C_q\}|$.

Consider the incidence graph H . If we draw the graph in the plane, we apply the crossing lemma. For any two distinct circles $C_p, C_q \in \mathcal{C}$, their intersection $|C_p \cap C_q|$ is at most 2. Therefore, there is no $K_{2,3}$ in the incidence graph. We apply the Kővári-Sós-Turán theorem explicitly. For

each point $p_i \in P$, let d_i be its degree (the number of circles it lies on). The number of pairs of circles containing a common point is $\sum_{i=1}^n \binom{d_i}{2}$. Since each pair of circles intersects at most twice, the maximum number of pairs of circles sharing a point across all points in P is bounded by $2\binom{n}{2} = n(n-1)$. Thus, $\sum_{i=1}^n \frac{d_i(d_i-1)}{2} \leq n(n-1)$. Using the Cauchy-Schwarz inequality, $\sum_{i=1}^n d_i^2 \geq \frac{1}{n}(\sum_{i=1}^n d_i)^2$. Therefore, $\frac{1}{2n}(\sum d_i)^2 - \frac{1}{2}\sum d_i \leq \sum \frac{d_i(d_i-1)}{2} \leq n^2$. Let $I = \sum d_i$. We have $\frac{I^2}{2n} - \frac{I}{2} \leq n^2$. Solving this quadratic inequality for I yields $I \leq \sqrt{2n^3} + \frac{n}{2}$. Thus, $u(P) = I/2 \leq \frac{1}{2}\sqrt{2n^3} + \frac{n}{4}$. This establishes the $O(n^{3/2})$ bound directly without omission. ■

5 Architecture for Autoformalization (Lean 4)

```
import Mathlib.Data.Real.Basic
import Mathlib.Analysis.InnerProductSpace.PiL2
import Mathlib.Combinatorics.SimpleGraph.Basic

/-!
# Architecture for the Erdős Unit Distance Problem
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-- We define points in the Euclidean plane.
abbrev Point := EuclideanSpace Real (Fin 2)

-- Definition of a unit distance
def is_unit_distance (p q : Point) : Prop :=
  dist p q = 1

-- Formalization of the unit distance graph
def UnitDistanceGraph (P : Finset Point) : SimpleGraph Point where
  Adj p q := p \in P /\ q \in P /\ p != q /\ is_unit_distance p q
  symm := by
    -- Symmetry of distance implies symmetry of adjacency
    intro p q h
    sorry
  loopless := by
    -- A point is distance 0 from itself, not 1
    intro p h
    sorry

-- The statement of the Erdős Conjecture
def ErdosUnitDistanceConjecture : Prop :=
  forall epsilon > 0, exists C : Real, C > 0 /\
  forall P : Finset Point,
  (UnitDistanceGraph P).edgeFinset.card <= C * (P.card : Real) ^ (1 + epsilon)

-- Lemma 2 statement
lemma point_circle_incidence_bound (P : Finset Point) :
  (UnitDistanceGraph P).edgeFinset.card <= (2 ^ (1/2)) * (P.card : Real) ^ (3/2) := by
  sorry
```